CMU-CS 462: Software Measurement and Analysis

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The basics of measurement

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Contents

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1. Metrology (Khoa học đo lường)

- Metrology is the science of measurement.
- Metrology is the basis for empirical science and engineering in order to bring knowledge under general laws, i.e., to distil observations into formal theories and express them mathematically.
- Measurement is used for formal (logical or mathematical, orderly and reliable) representation of observation.

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Two Problem Categories

- Components of a measurement system:

  m = <*attribute, scale, unit*>

  - Attribute is what is being measured (e.g., size of a program)
  - Scale is the standard and scope of measurement (e.g., nominal, ordinal, ratio scale, etc.)
  - Unit is the physical meaning of scale (e.g., a positive integer, a symbol, etc.)
- Determining the value of an attribute of an entity.
- Determining the class of entities to which the measurement relates.

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Empirical Relations

- Empirical relation preserved under measurement M as numerical relation

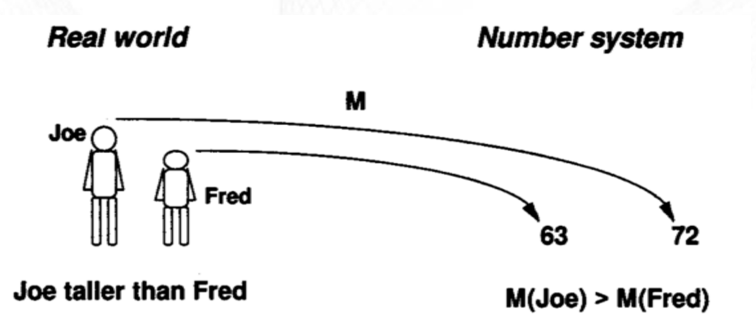

Figure from Fenton's Book

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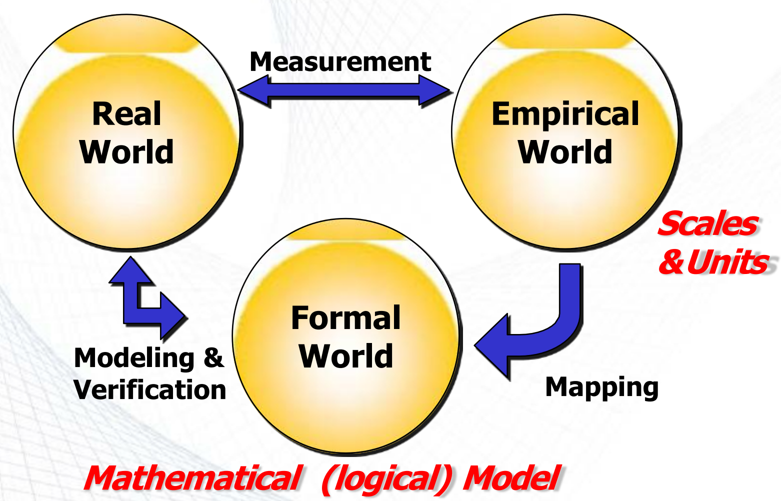
Real, Empirical & Formal Worlds



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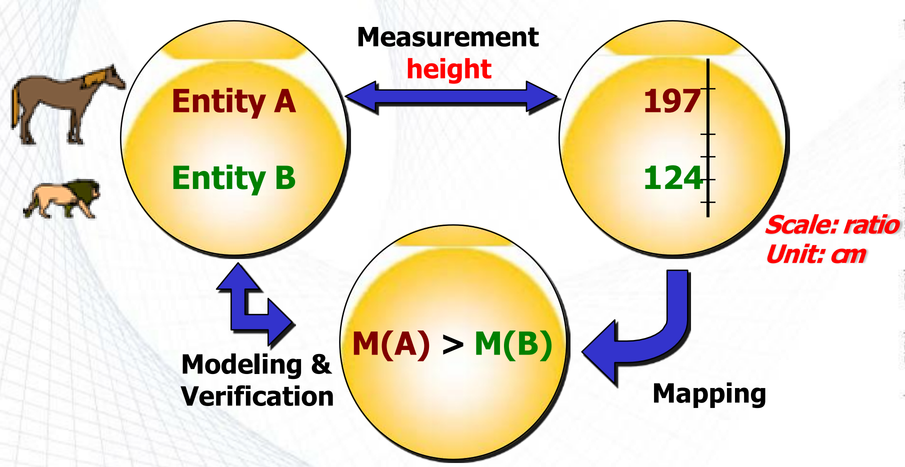
Real, Empirical & Formal Worlds



Measurement: Activities /1

- **Problem definition**
  - Defining measurement problem
  - Designating set of entities forming the target of measurement
  - Identifying key attributes for the entities
- **Identifying scales**
  - Identifying scales for which the attributes can be measured
- **Forming the empirical relational system**
  - Mapping entities and their selected attributes to numbers or values on the scales

Measurement: Activities /2

- **Modeling**
  - Developing mathematical (logical) representation of the entities and their attributes
- **Defining the formal relational system**
  - Mapping the empirical relational system to the formal model
- **Verifying the results of measurement**
  - Verifying whether the measurement results reflect the properties of the entities, attributes and relationships

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Empirical Relational System

- $E = \{\underline{A}, R, O\}$
- $A = \{a, b, c, ..., z\}$ set of real world entities
- k: property for each element of A
- $\underline{A} = \{\underline{a}, \underline{b}, \underline{c}, ..., \underline{z}\}$ is the model of A which describes each element of A in terms of the property k
- Empirical observations defined on the model set $\underline{A}$
  - Set of n-ary relations: R, e.g., X is taller than Y
  - Set of binary operations: O , e.g., X is tall

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Formal Relational System

- $F = \{\underline{A}', R', O'\}$
- F should satisfy the following conditions:
  - Capable of expressing all relations and operations in E.
  - Support the meaningful conclusions from the data.
  - Mapping from E to F must represent all the observations, preserving all the relations and operations of E.

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Example 1 /1

Ranking 4 software products based on user preferences

- Problem definition: ranking the products (entities) A, B, C and D based on user preferences
- Scale: A single 0-100% linear scale
- Empirical relational system: represented by the table.

	A	B	C	D
A	-	80	10	80
B	20	-	5	50
C	90	95	-	96
D	20	50	4	-

(A,B) = 80 means %80 of the users preferred product A to B

Example from Fenton's Book

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Example 1 /2

- Modeling:
  - Valid pairs are those having the value greater that 60.
  - If for a pair (A,B) , more than 60% of users prefer A to B then A is "definitely better" than B.

    If M(x,y) > 60% then p(x) > p(y)
- Formal relation system:

  If p(x) > p(y) and p(y) > p(z)

  Then p(x) >p(z)
- Verification:
  - Valid pairs (C,A), (C,B), (C,D), (A,B) and (A,D)
  - No conflict between the data collected and the model

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Example 1 /3

- Verification fails if the data was collected as shown here because

M(B,C) > 60% then p(B) > p(C)
M(C,A) > 60% then p(C) > p(A)
M(A,B) > 60% then p(A) > p(B)

- There is a conflict between the real and formal model and the model must be revised.

	A	B	C	D
A	-	80	10	80
B	20	-	95	50
C	90	5	-	96
D	20	50	4	-

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Example 2 /1

- Entity: software failure
- Attribute: criticality
- Three types of failure is observed:
  - Delayed response
  - Incorrect output
  - Data loss
- There are 3 failure classes in E
  - Delayed response (R1)
  - Incorrect output (R2)
  - Data loss (R3)

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Example 2 /2

- Measurement mapping

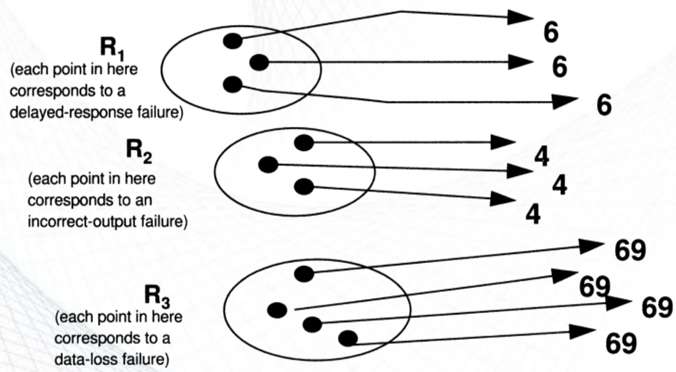

Figure from Fenton's Book

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Example 2 /3

- Next we would like to add a new binary relation:
  x is more critical than y
- Each data loss failure (x in R3) is more critical than incorrect output failure (y in R2) and delayed response failure (y in R1)
- Each incorrect output failure (x in R2) is more critical than delayed response failure (y in R1)
- The model should be revised to account for this new binary relation

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Example 2 /4

- Revised measurement mapping

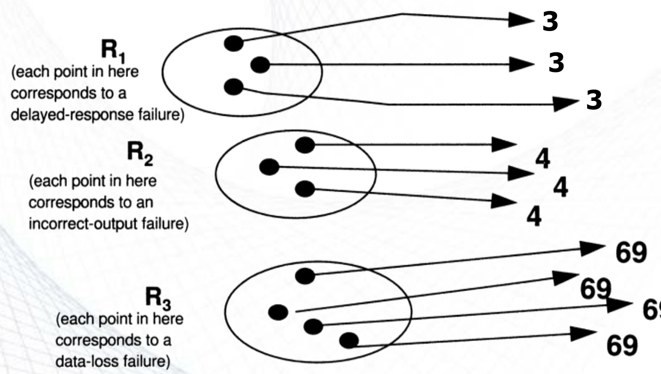

Figure from Fenton's Book

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Measurement: Properties /1

- **Completeness**
  - Components of a measurement system:
    $m = <attribute, scale, unit>$
  - Attribute is what is being measured (e.g., size of a program)
  - Scale is the standard and scope of measurement (e.g., ratio)
  - Unit is the physical meaning of scale (e.g., a positive integer)
- **Uniqueness**
  - A measurement result should be unique and match with the scale and units

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Measurement: Properties /2

- **Extendibility**
  - Two or more formal measures mapping to the same entity are compatible using explicit compatibility relation (e.g., *cm* and *inch* used to measure length and $1\ in = 2.54\ cm$)
- **Discrete Differentiability**
  - The minimum unit of measurement scale is used to determine the differential rate of the measurement system.

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Measurement: Properties /3

- **Deterministic or Probabilistic**
  - Measurement system should be either deterministic (i.e., lead to same results under same conditions) or probabilistic (e.g., productivity per hour, reliability)
- **Quantitative or Qualitative**
  - Result of measurement is either quantitative (represented by number values) or qualitative (represented by qualitative values or range intervals)

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Measurement: Direct & Indirect

- Direct measurement of an attribute of an entity involves no other attribute or entity.
  - E.g., software length in terms of lines of code.
- Indirect measurement is useful in making visible the interaction between direct measurements.
  - E.g., productivity, software quality.
- Indirect measurement is particularly helpful when the size of empirical measures is quite big (i.e., many relations and many entities) or the cost of direct measurement is high.

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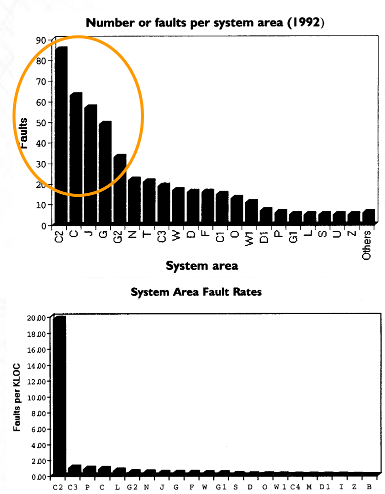
Example: Direct & Indirect

- In a software system, measuring number of faults (i.e., direct measurement) leads to identification of 5 problem areas.
- However, measuring faults per KLOC (i.e., indirect measurement) leads to identification of only one problem area.



Example from Fenton's Book

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Measurement: Size of Empirical Set

- **Example:** measuring popularity of 4 software products
- $E = \{\underline{A}, R, O\}$
  - |A| = 4
  - |R| = 3
    - r1: x is more popular than y
    - r2: x is less popular than y
    - r3: indifferent
- The total number of possible measures is:

  $4 \times (4 - 1) \times 3 = 36$
- The size will grow when the number of elements and relations grow.

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Readings and Assignment 2

- You are required to read the video in the link below and Answer the following questions. The answers will be submitted on e-Learning system
  - https://www.youtube.com/watch?v=uBEZoXc4A5w
  - Question 1
    - What are Flow Metrics in Software Delivery Value Streams?
    - Who long does it take for the team complete work, from acceptance to release?

Software Measurement!

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